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MKR Vidor 4000

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This tutorial refers to a
product that has reached
its end-of-life status.

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QR Code Recognition

The QR library allows you to recognize
QR code markers and data.

Author • Arduino Last
revision · 12/19/2023

This example shows the flexibility
and power of the FPGA
architecture. A QR code
recognition set of APIs is included
in the VidorGraphics library and
the corresponding IP is in the
bitstream that is loaded in the
FPGA together with the sketch for
the SAMD Board Package. The
example highlights on video the
QR code markers and prints on
the serial monitor the coordinates
of the markers found.

Hardware Required

Arduino MKR Vidor 4000

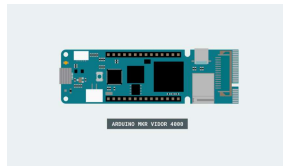
Omnivision OV5647 camera

microHDMI to HDMI cable
or adaptor

Monitor with HDMI input

Help

There is no circuit for this example.



The circuit for this tutorial.

Code

Include the VidorCamera library, which is part of VidorGraphics.

```
#include "VidorGraphics.h"
```

```
#include "VidorCamera.h"
```

You have a number of functions available to manage the recognition of a QR Code. VidorQR is part of the VidorCam library; the functions are accessible with **cam.qrrec**.

```
void begin()
```

 - start recognition algorithm

```
void end()
```

 - stops recognition

```
void setMode(uint8_t mode);
```

- change video mode

```
void setThr(uint8_t thr);
```

- set recognition threshold

```
int readQRCode(void);
```

 -

perform qr code recognition

```
sQrDet qr;
```

 - structure

containing detected points

```
1 #include "VidorGraph
2 #include "VidorCamer
3
4 VidorCamera vcam;
5
6 #define MAXDIM 10
7
8 static uint16_t x[QR
9
10 struct qrPtn {
11     uint16_t x[QR_PT_D
12     uint16_t y[QR_PT_D
13 };
14
15 static qrPtn qrBufFe
16
17 uint16_t count = 0,
18
19 void setup() {
20     Serial.begin(9600)
21     // wait for the se
22     // if you are powe
23
24
25
26
27
28
```

Last revision 2018/07/22 by SM

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